CA Lab Exam - Section 3MCA06 - Set A

College Student Records

Your task

The web page and the Express server are ready. The database part is missing. Complete it using Mongoose.

- Edit only **database.js**. The comments in it tell you what to write.
- Do not change any other file.

Steps

1. Open this project folder in VS Code and open a terminal in it.
2. Install the packages:

```
npm install
npm install mongoose
```

3. Make sure MongoDB is running, then add the sample data (see below).
4. Complete **database.js**.
5. Start the server:

```
node index.js
```

6. Open **http://localhost:3000** in your browser.

After every change to database.js, stop the server with Ctrl + C and start it again.

Database details

Database: college

Collection: students

Fields: name (text), dept (text), city (text), marks (number)

Sample data

Open mongosh, paste these commands and press Enter. Paste them only once.

```
use college
db.students.insertMany([
  { name: "Aarav", dept: "CSE", city: "Bengaluru", marks: 88 },
  { name: "Diya", dept: "ECE", city: "Mysuru", marks: 76 },
  { name: "Rohan", dept: "MECH", city: "Hubballi", marks: 64 },
  { name: "Kavya", dept: "CSE", city: "Mangaluru", marks: 95 },
  { name: "Ishaan", dept: "ECE", city: "Bengaluru", marks: 82 },
  { name: "Meera", dept: "MECH", city: "Mysuru", marks: 71 },
  { name: "Arjun", dept: "CSE", city: "Hubballi", marks: 79 },
  { name: "Sneha", dept: "ECE", city: "Mangaluru", marks: 91 },
  { name: "Vikram", dept: "MECH", city: "Bengaluru", marks: 58 },
  { name: "Ananya", dept: "CSE", city: "Mysuru", marks: 86 },
  { name: "Rahul", dept: "ECE", city: "Hubballi", marks: 69 },
  { name: "Priya", dept: "MECH", city: "Mangaluru", marks: 93 }
])
```

Show this output to the examiner

1. Terminal. Run node index.js.

Expected: Connected to MongoDB, then Server running at http://localhost:3000

2. Show all. Click "Show All Students".

Expected: 12 students.

3. Search. Choose CSE and click Search.

Expected: 4 students: Aarav, Kavya, Arjun, Ananya.

4. Filter. Type 85 in "Marks at least" and click Filter.

Expected: 5 students: Aarav, Kavya, Sneha, Ananya, Priya.

5. Top 3 Students. Click "Top 3 Students".

Expected: Kavya (95), Priya (93), Sneha (91), in this order.

6. Add Student. Fill the form with Name: your own name, Department: CSE, City: Bengaluru, Marks: 80. Click "Add Student".

Expected: The message "Student saved" appears and the list shows 13 students, including your name.